University of Illinois Urbana-Champaign
Information Science (PhD), Fall 2025

Pratyasha Saha
pratyasha.hci@gmail.com | www.sites.google.com/view/pratyashasaha

Academic Statement of Purpose

My academic and career objectives are focused on making a significant contribution to the area of inclusive and responsible technology design that prioritizes ethics, privacy, justice, and equitable access, especially for marginalized communities. My goal is to become a professor or academic researcher who analyzes and designs technology through a critical lens of social responsibility. To date, I have co-authored 18 HCI publications, including 10 in CHI and CSCW, and a Best Paper Award at CHI 2024 as the first author.

Over the past six years, my passion for research has grown through collaborations with leading HCI scholars. My work has given me firsthand insights into how technological advancements, despite their potential social benefits, can also amplify vulnerabilities, reinforce biases, and create barriers for marginalized communities. These experiences have inspired me to pursue a Ph.D. in Information Science where I can deepen my understanding of these challenges and craft informed solutions.

Privacy for the Marginalized. I first got the opportunity to explore the field of HCI in 2016 through a collaborative project led by Professors Ishtiaque Ahmed, Nicola Dell and Jay Chen. I volunteered to conduct 25 semi-structured interviews at urban slums to understand how people share phones among family members, and what privacy challenges they face. I found that privacy perceptions in these communities are rooted in their cultural values of sharing, which differ sharply from Western contexts. In 2018, I collaborated again with Prof. Ahmed and Prof. Taiabul Haque to explore traditional community practices to exchange confidential and sensitive information. I conducted a three-month-long ethnography with more than 80 individuals of Hijra (transgender), Harijan, Bede (snake-charmers), and indigenous communities. My work on this project led to a publication in CSCW 2019 [10], which further solidified my commitment to culturally and contextually relevant HCI research.

Gender, Identity, and Justice. In 2017, I started working with Prof. Shion Guha to understand how gender-based marginalized groups construct identity and combat harassment on social media. I explored literature on feminist, cultural, and intersectionality theories and collected data from sexual harassment survivors. My findings revealed that women were facing more harassment in anonymous social platforms like Sarahah and ask.fm, which silenced their voices in these platforms and even led to non-use. I also felt the need to expand the work and proposed my collaborators to include the Hijra and LGBTQIA+ communities. I spent a year reaching out to the community people through personal connections, building trust, and proving that I could be a safe space for them. Additionally, I developed skills in designing surveys in Qualtrics, performing R-based quantitative analysis, and applying qualitative methods such as open coding, thematic analysis, and grounded theory. The three-year journey led to six publications [2-7] in top HCI venues like CSCW and ICTD and, more importantly, left me with a profound sense of empathy. In these papers, we argued for more inclusive technology design with appropriate content moderation policies that support gender-justice and equity.

Trust, Stigma, and Surveillance. In 2021, I independently designed a study on how one of the most marginalized communities in the world – sex workers in red-light districts – navigate online spaces amidst stigma, moral tension, and surveillance. I explored literature around digital intimacy, sex work and technology, Sassen's spatial theory, and Durkheim's theory of sacred and profane. For three long months, I regularly traveled more than 180 miles to the field to build rapport until I became a familiar and trustworthy face there. My ethnographic findings revealed how sex workers in this part of the world use digital platforms for their work, build trust in these systems, and seek help with tech-related challenges. This work was recognized by a Best Paper Award at CHI this year [1].

Community-centered Technology Design. To gain experience on translating my theoretical understanding into design, I joined Prof. Patrick Olivier, and Dr. Bronwyn Cumbo at Monash University as a Research Assistant (remote) for the PROTIC II project in Spring 2021. My first work under this project was to improve an existing maid-booking gig-platform for domestic workers in Dhaka. I conducted participatory workshops, semi-structured interviews, and spent hours in co-designing solutions that balanced the needs of both workers and employers. Analyzing our findings, we integrated an IVR framework into the existing system which helped workers interact with the platform via phone calls. We separately designed Paroli, a telecom platform now used in Nepal, Australia, and by Oxfam in Bangladesh for community engagement. This experience taught me the power of co-design and community input in solving complex socio-technical problems. Our work on the IVR framework was published in DIS 2023 [9], and another study on Paroli users' data privacy perceptions resulted in a publication in CHI 2024 [8].

Currently, I'm interested in exploring emerging technologies and their implications in marginalized and vulnerable contexts around privacy and ethics. I also want to design accessible and equitable systems using AI methodologies. At UIUC, I'm particularly interested in working with Prof. **Anita Say Chan** whose research around *data ethics and cultures,* and *feminist and decolonial approaches to technology* resonates best with my interests. I'm also highly interested in working with Prof. **Michael Twidale** to study *metaphors, mental models and misconceptions in learning new technologies (VUI, XR/AR),* and *social aspects of ICT use and their implications for design*. I'm also eager to work with Prof. **Yue Guo** to *enhance the accessibility and personalization of health information* for laypeople *through NLP methodologies.*

Throughout my work, I have identified a critical need for scholarship that explores the intersection of marginality and technology, particularly in assessing how people's values around culture, faith, and traditions influence their interaction with digital technologies. As a Ph.D. student at UIUC, I aim to deepen this research by studying the ways socio-technical systems affect marginalized communities and by developing frameworks for culturally informed technology design to empower them. I believe pursuing my PhD in a highly interdisciplinary program like UIUC Information Science will help me achieve my PhD goals, and grow as a strong, well-rounded researcher.

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